Deep Learning Notes

Instructor

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1 Introduction to Deep Learning

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers.

2 Neuron Model

A neuron computes:

$$z = w^T x + b$$

$$a = \sigma(z)$$

Where:

- x = input vector
- w = weights
- b = bias
- σ = activation function

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3 Activation Functions

3.1 Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

3.2 ReLU

$$f(x) = \max(0, x)$$

3.3 Tanh

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

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4 Loss Functions

4.1 Mean Squared Error

$$L = \frac{1}{n} \sum (y - \hat{y})^2$$

4.2 Cross Entropy

$$L = -\sum y \log(\hat{y})$$

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5 Gradient Descent

$$\theta = \theta - \eta \nabla L$$

Where:

- η = learning rate
- ∇L = gradient

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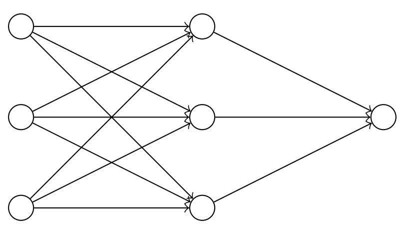
6 Neural Network Diagram



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7 Forward Propagation

$$Z^{[l]} = W^{[l]} A^{[l-1]} + b^{[l]}$$

$$A^{[l]} = \sigma(Z^{[l]})$$

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8 Backpropagation

$$dZ = A - Y$$

$$dW = \frac{1}{m} dZ \cdot A^T$$

$$db = \frac{1}{m} \sum dZ$$

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9 Python Example (Neural Network)

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import numpy as np

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

X = np.array([[0,0],[0,1],[1,0],[1,1]])
y = np.array([[0],[1],[1],[0]])

W = np.random.rand(2,1)
b = 0

for i in range(1000):
    z = np.dot(X, W) + b
    a = sigmoid(z)
    loss = np.mean((y - a)**2)

print(a)
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10 Convolutional Neural Networks (CNN)

$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n I(m,n)K(i-m, j-n)$$

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11 Recurrent Neural Networks (RNN)

$$h_t = \tanh(W_h h_{t-1} + W_x x_t)$$

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12 Conclusion

Deep Learning models learn hierarchical representations using neural networks and optimization techniques.